



**Academia Internacional David
Robotics and Technology
2nd Trimester
Summative # 1
8th A & B**

Name: _____ Date: _____

Teacher: Porfirio Rios Group: 8° A B Score: ____ / 40pts

Arduino Basics Vocabulary Table

Criteria	9	7	5	2
Vocabulary completion and definitions	All 10 words have accurate, student-friendly definitions.	Most definitions are accurate.	Some definitions are missing or unclear.	Definitions are mostly missing or incorrect.
Illustrations or diagrams	Each word has a relevant illustration or diagram.	Most illustrations match the words.	Some illustrations are missing or unclear.	Illustrations are mostly missing or unrelated.
Example sentences	Sentences use each word in a clear Arduino or Freenove context.	Most sentences use the words correctly.	Some sentences are incomplete or vague.	Sentences are mostly missing or incorrect.
Hardware accuracy	Board, circuit, breadboard, pin, power, ground, resistor, and LED ideas are correct.	Most hardware ideas are correct.	Some hardware ideas are confused.	Hardware ideas are mostly incorrect.
Criteria	4	3	2	0
Punctuality, Readiness & Respect	Files were delivered on time to Google Classroom, the student was prepared to work, and had necessary resources, e.g. computer, charger, email, and password.	Files were delivered on time, with only one minor readiness or resource issue.	Files were submitted, but there was a noticeable lateness, readiness, or resource problem.	Files were not delivered on time to Google Classroom, the student was not prepared to work, or did not have the necessary resources, e.g. computer, charger, email, and password.

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Button-Controlled LED Build

Criteria	9	7	5	2
Working circuit	Builds or demonstrates a button-controlled LED circuit that changes state when pressed.	Circuit works mostly as expected, with only one minor wiring, upload, or button-response issue.	Circuit partly works, but the LED or button response is inconsistent.	Circuit does not work or the button does not control the LED.
Labeled evidence	Submits a clear labeled photo or diagram of the circuit.	Submits labeled evidence with only one minor missing or unclear label.	Submits evidence, but some labels or parts are missing.	Photo/diagram evidence is missing or not understandable.
Input/ process/ output explanation	Explains the input, process, and output correctly in one sentence.	Explains the input, process, and output with only one minor unclear detail.	Explanation includes some system parts, but one part is missing or unclear.	Explanation is missing or does not match the circuit.
Testing evidence	Tests safely and records three button/LED observations or check results.	Tests safely and records observations, with only one minor missing or unclear note.	Tests the circuit, but observations are incomplete or need reminders.	Testing evidence is missing, or materials are handled unsafely.
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Potentiometer Data Chart

Criteria	9	7	5	2
Data readings	Includes at least 6 potentiometer positions with analog readings.	Includes the required readings, with only one minor unclear or missing value.	Includes some readings, but fewer than 6 or some values are unclear.	Readings are mostly missing or not usable.
Brightness observations	Records LED brightness observations for the data points.	Records brightness observations for most data points, with only one minor gap.	Records some brightness observations, but they are incomplete or vague.	Brightness observations are missing or not connected to readings.
Chart organization	Data chart is labeled, neat, and easy to compare.	Chart is mostly labeled and organized, with only one minor issue.	Chart can be understood but has some missing labels or organization issues.	Chart is difficult to read or not organized as a data chart.
Conclusion	Writes one clear conclusion about the relationship between input and output.	Writes a conclusion mostly supported by data, with only one minor missing detail.	Writes a conclusion, but it is too general or only partly supported by data.	Conclusion is missing, incorrect, or not based on the data.
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Servo or Buzzer Input-Control Build

Criteria	9	7	5	2
Controlled output	Builds or demonstrates a servo or buzzer system controlled by a button, potentiometer, or sensor.	System mostly works, with only one minor input or output-response issue.	System partly works, but the input or output response is inconsistent.	System does not show a working input-controlled output.
Labeled diagram	Submits a labeled diagram showing the input, selected output, and main circuit connections.	Submits a labeled diagram with only one minor missing or unclear label.	Submits a diagram, but some labels or connections are missing.	Diagram is missing or not understandable.
Input and output behavior	Clearly identifies the input used and the output behavior produced.	Identifies the input and output behavior, with only one minor unclear detail.	Identifies input/output, but the behavior detail is unclear.	Input/output evidence is missing or does not match the build.
Explanation	Explains how the input controls the selected output using simple physical computing language.	Explains how the input controls the output with only one minor missing detail.	Explanation is partly correct but too general or missing one important detail.	Explanation is missing or does not match the system.
Criteria	4	3	2	0
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Arduino Project Design Plan

Criteria	9	7	5	2
Problem and role	Plan states the problem/purpose and identifies the individual or pair role.	Problem/purpose and role are mostly clear, with only one minor missing detail.	Plan includes a problem or role, but one part is incomplete or unclear.	Problem/purpose and role are mostly missing.
Input and output components	Identifies one input component and one output component that fit the project.	Identifies fitting input/output components, with only one minor unclear connection.	Identifies components, but one choice is unclear or weakly connected.	Input/output components are missing or do not fit the project.
Circuit diagram and behavior	Includes a circuit diagram and expected behavior that can guide building.	Diagram and expected behavior mostly guide building, with one minor missing detail.	Includes a diagram or behavior description, but one part lacks enough detail.	Diagram and expected behavior are missing or not usable.
Testing and materials	Includes 3 test cases and a realistic materials list.	Includes test cases and materials, with only one minor incomplete detail.	Includes some tests or materials, but one requirement is incomplete.	Testing plan and materials list are mostly missing.
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